

In this project I aim to estimate the effect of sitting NBA players out for back to back games on the probability of getting injured and the average severity of injury. In the past decade, it has become popular for NBA teams to sit players in one of a pair of back to back games in order to reduce the likelihood that the player gets injured, particularly for older, high usage players. Because the distribution of the treatment is so uneven, this makes it a good opportunity to apply the methods we learned in class, particularly propensity scores and inverse probability weighting, to estimate the average treatment effect. The data will be obtained mostly through the code from the airball R package (<https://github.com/josedv82/airball/tree/master>) which can be used to scrape schedule, travel, and injury information. Additional data will be obtained from Basketball Reference in order to use player characteristics and statistics as additional covariates. Since the data has not been pieced together and it will take substantial effort to do so, I can not yet provide figures for the number of observations. The NBA's official website says that the average NBA team played 14 back to back sets per year, so the total number of observations should be around $14 * 30 \text{ teams} * 8\text{-}12 \text{ rotation players per team} * 10 \text{ years} = 33,600 - 50,400$ observations. Below is a table of the covariates I intend to include.

Covariate	Description	Type
Held out of half of B2B (treatment)	Whether a player was held out of half of a back to back set.	Binary
Injury (outcome)	Whether a player was reported as having an injury within 3 days of the back to back set.	Binary
Injury Severity	The length of time a player was out due to an injury suffered within 3 days of a back to back set. 0 if no injury occurred.	Continuous
3in4	Whether the second game of the back to back set was the 3rd game played in 4 days.	Binary
4in5	Whether the second game of the back to back set was the 4th game played in 5 days.	Binary
5in7	Whether the second game of the back to back set was the 5th game played in 7 days.	Binary
Age	Age of player	Continuous

MPG	Minutes per game of player in season in question	Continuous
USG	Usage rate of player in season in question	Proportion
Cumulative Minutes	Cumulative minutes of the player in question for his entire career	Continuous
Career USG	Usage rate of player in question for his entire career	Proportion
Travel G1	Whether the player had to travel by plane to get to the first game of the back to back	Binary
Travel G2	Whether the player had to travel by plane to get to the second game of the back to back	Binary
Distance Traveled	Total distance traveled in miles by plane for the player to get to the two games of the back to back	Continuous